

ArabBot Studio

AI Chatbot Platform for Egyptian SMBs

WhatsApp Business · Egyptian Arabic · RAG Knowledge Base

Generated from README.md

build & deploy AI chatbots for WhatsApp Business —

understands *العامة المصرية*, connects to your business tools, works 24/7.</i>

Features

- **Egyptian Arabic AI** — Understands *العامة المصرية*, not just MSA
- **WhatsApp Business API** — One-click connect, webhook verified
- **9 Intent Handlers** — Greeting, FAQ, Order, Complaint, Human Handoff, and more
- **RAG Knowledge Base** — Upload FAQs, AI answers with your data (FAISS vector search)
- **Human Handoff** — Escalate to a real person when the bot can't help
- **Workspace Isolation** — Multi-tenant, each customer's data fully isolated
- **JWT Auth** — Secure API access with role-based workspace membership
- **Analytics Dashboard** — Message volume, intent distribution, response times
- **Warm Constructivist UI** — Distinctive design system with terracotta/navy palette

Tech Stack

Layer	Technology
Backend Framework	FastAPI (async Python)
Frontend Framework	React 19 + TypeScript ~6.0
Build Tool	Vite 8
CSS	Tailwind CSS 4
Charts	Recharts
AI / LLM	LangChain 0.3 + Google Gemini
Database	PostgreSQL 16 (async with asyn
Vector Store	FAISS (local embedding search)
Cache / Queue	Redis (background task queue)
Auth	JWT (python-jose) + bcrypt
Migrations	Alembic

AI Pipeline

Project Structure

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```

├── unit/
│   └── test_chains/      # 4 tests
├── alembic/              # DB migrations
│   ├── env.py
│   ├── script.py.mako
│   └── versions/001_initial.py
├── alembic.ini
├── requirements.txt
├── .env.example
├── Dockerfile
├── docker-compose.yml    # PostgreSQL 16 + Redis 7 + app
├── frontend/
│   ├── src/
│   │   ├── main.tsx      # Entry point (React StrictMode)
│   │   ├── App.tsx       # Router + QueryClient + AuthProvider
│   │   ├── index.css     # Tailwind 4 + Warm Constructivist design system
│   │   ├── App.css       # Legacy styles
│   │   └── pages/        # 10 page components
│   │       ├── Login.tsx  # Split-screen login form
│   │       ├── Register.tsx # Split-screen registration
│   │       ├── Dashboard.tsx # Stat cards + recent bots
│   │       ├── BotsList.tsx # Bot management table
│   │       ├── BotEditor.tsx # Create/edit bot config form
│   │       ├── KnowledgeBase.tsx # FAQ items per bot
│   │       ├── Conversations.tsx # Conversation viewer + message thread
│   │       ├── Analytics.tsx # Recharts dashboard
│   │       ├── Handoffs.tsx # Human handoff queue
│   │       └── Settings.tsx # User profile + webhook guide
│   │   ├── components/   # Shared UI components
│   │   │   ├── Layout.tsx # Auth-gated layout with sidebar
│   │   │   ├── Sidebar.tsx # Navigation sidebar
│   │   │   └── LoadingSpinner.tsx # Terracotta spinner
│   │   └── lib/
│   │       ├── api.ts     # Axios client + JWT interceptor (16 functions)
│   │       └── auth.ts     # Auth context provider
│   │   └── types/
│   │       └── index.ts    # TS interfaces (10 types)
│   ├── index.html        # Google Fonts (Space Grotesk + DM Sans)
│   ├── vite.config.ts     # Proxy /api + /webhooks → :8000
│   ├── tsconfig.json
│   ├── tsconfig.app.json
│   ├── tsconfig.node.json
│   ├── eslint.config.js
│   └── package.json
├── AGENTS.md             # Agent instructions for AI assistants
├── specs/                # Feature specifications
│   └── 001-arabbot-mvp/   # MVP spec, plan, research, data model, contracts
├── features/             # Feature definitions
└── README.md

```

AI Pipeline

The chatbot message flow processes incoming WhatsApp messages through 5 sequential stages:

Stage	File	Model	Purpose
1. Dialect Normalizer	`chains/dialect_normalizer.py`	Gemini 2.0 Flash	Converts Egyptian Arabic slang
2. Intent Router	`chains/intent_router.py`	Gemini 2.0 Flash	Classifies into 9 intents: GRE
3. Intent Handlers	`chains/intent_handlers.py`	—	Predefined Arabic responses fo
4. RAG Engine	`chains/rag_chain.py` + `servi	Gemini 2.5 Pro	FAISS vector search → context-
5. Orchestrator	`chains/orchestrator.py`	—	Coordinates stages 1-4, decide

Intent Routing Logic

```

GREETING      → random Arabic greeting (no AI call)
HUMAN_REQUEST → Arabic fallback + create handoff ticket
COMPLAINT     → Arabic apology + create handoff ticket
OTHER/INTENTS → knowledge search → RAG generation (Gemini 2.5 Pro) → reply

```

Quick Start

Prerequisites

- Python 3.12+
- Node.js 20+
- PostgreSQL 16+ (or SQLite for local dev)
- Redis 7+ (optional)
- Google Gemini API key (for AI features)

1. Clone & Setup Backend

```

git clone https://github.com/AHMEDGabal1/arabbot-studio.git
cd arabbot-studio/backend

python -m venv .venv
# Windows: .venv\Scripts\activate
# Linux/Mac: source .venv/bin/activate

pip install -r requirements.txt

```

2. Configure Environment

```
cp .env.example .env
```

Edit `.env` with your credentials:

Variable	Description
<code>`DATABASE_URL`</code>	<code>`postgresql+asyncpg://user:pas`</code>
<code>`GOOGLE_API_KEY`</code>	Your Google Gemini API key
<code>`SECRET_KEY`</code>	Random 32+ character string fo
<code>`ENVIRONMENT`</code>	<code>`development`</code> or <code>`production`</code>

3. Setup Database

```

# For PostgreSQL:
alembic upgrade head

# For SQLite (dev only):
python -c "import asyncio; from sqlalchemy.ext.asyncio import create_async_engine; from src.database import Base; import src.models; e = create_async_engine

```

4. Start Backend Server

```
python -m uvicorn src.main:app --reload --port 8000
```

Backend runs at ``http://localhost:8000`` — API docs at ``http://localhost:8000/docs``

5. Setup Frontend

```
cd arabbot-studio/frontend
npm install

```

```
npm run dev
```

Frontend runs at `http://localhost:5173` (proxied to backend at `:8000`)

6. Verify It's Live

```
curl http://localhost:8000/health
# → {"status":"ok","database":"up","redis":"unknown"}
```

API Reference

Authentication

Method	Endpoint	Request Body	Description
POST	/api/v1/auth/register	{ email, password, name }	Register user + create workspa
POST	/api/v1/auth/login	{ email, password }	Login → JWT (with `workspace_i
POST	/api/v1/auth/refresh	{ refresh_token }	Refresh JWT token

| GET | /api/v1/auth/me | — | Current user profile (requires `Authorization: Bearer <token>`)

All authenticated endpoints require `Authorization: Bearer <jwt\_token>` header. JWT contains `sub` (user_id) and `workspace\_id` claims. Expires in 24 hours.

Bots

Method	Endpoint	Description
GET	/api/v1/bots	List workspace bots
POST	/api/v1/bots	Create a bot
GET	/api/v1/bots/{id}	Get bot details
PATCH	/api/v1/bots/{id}	Update bot
DELETE	/api/v1/bots/{id}	Delete bot (soft)
POST	/api/v1/bots/{id}/activate	Activate bot
POST	/api/v1/bots/{id}/deactivate	Deactivate bot

Knowledge Base

Method	Endpoint	Description
GET	/api/v1/bots/{bot_id}/knowled	List knowledge items
POST	/api/v1/bots/{bot_id}/knowled	Add knowledge item + add to FA
DELETE	/api/v1/bots/{bot_id}/knowled	Delete item from DB (FAISS rem
POST	/api/v1/bots/{bot_id}/knowled	Rebuild FAISS index from DB it

Conversations

Method	Endpoint	Query Params	Description
GET	/api/v1/conversations	bot_id, status, limit (1-	List conversations (workspace-
GET	/api/v1/conversations/{id}	—	Get single conversation
GET	/api/v1/conversations/{id}/me	—	Get conversation messages

Analytics

Method	Endpoint	Description
GET	/api/v1/analytics/overview	Workspace-level analytics (bot
GET	/api/v1/analytics/bots/{bot_i	Per-bot analytics (conversatio

Webhooks (WhatsApp)

Method	Endpoint	Description
GET	/webhooks/whatsapp/{bot_id}	Meta verification challenge (
POST	/webhooks/whatsapp/{bot_id}	Incoming WhatsApp messages (HM

Handoffs

Method	Endpoint	Description
GET	/api/v1/handoffs	List pending (unresolved) hand
PATCH	/api/v1/handoffs/{id}/assign	Assign handoff to an agent
PATCH	/api/v1/handoffs/{id}/resolve	Resolve handoff

System

Method	Endpoint	Description
GET	/health	Health check (DB status + Redi

Data Model

8 tables in the database (simplified relationships):

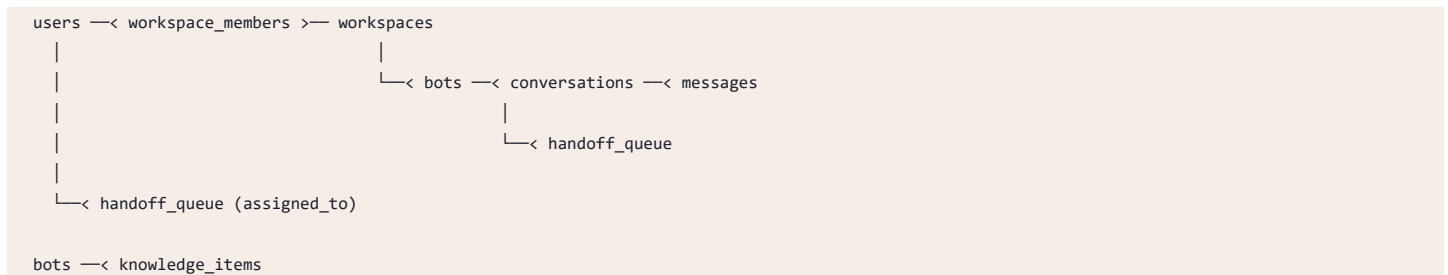


Table	Key Columns	Purpose
users	id (UUID PK), email (unique)	Authentication
workspaces	id (UUID PK), plan, month	Multi-tenant isolation
workspace_members	workspace_id + user_id (composite PK)	User-workspace membership
bots	id (UUID PK), workspace_id (FK)	Chatbot configuration
conversations	id (UUID PK), bot_id (FK)	Per-user chat sessions
messages	id (UUID PK), conversation_id (FK)	Message history + AI metadata
knowledge_items	id (UUID PK), bot_id (FK)	FAQ knowledge base
handoff_queue	id (UUID PK), conversation_id (FK)	Human escalation tickets

WhatsApp Webhook Setup

- Create a bot via `POST /api/v1/bots` with your WhatsApp Business `phone_number_id` and `access_token`
- In Meta Developer Portal → WhatsApp → Configuration, set:
 - Callback URL**: `https://your-domain.com/webhooks/whatsapp/{bot_id}`
 - Verify Token**: any string (not validated yet, only `hub.challenge` is echoed)
- Subscribe to `messages` webhook field
- Backend verifies incoming messages via HMAC-SHA256 using the bot's `wa_access_token` as the secret

Message Flow

WhatsApp → POST /webhooks/whatsapp/{bot_id}

- HMAC verify (X-Hub-Signature-256)
- Parse payload (text/interactive messages)
- Background task: get/create conversation
- Save user message
- FAISS knowledge search
- AI pipeline (normalize → classify → route → respond)
- Save assistant message
- Send reply via Meta Cloud API
- (Optional) Create handoff ticket

Frontend Design System

The dashboard uses a Warm Constructivist aesthetic — a distinctive alternative to generic SaaS UIs.

Token	Value	Usage
<code>--color-navy-700</code>	<code>#1a1f2e</code>	Sidebar, primary buttons
<code>--color-terracotta-500</code>	<code>#c1694f</code>	Accent, active states
<code>--color-gold-400</code>	<code>#e9b741</code>	Secondary accent
<code>--color-sand-100</code>	<code>#f5ede6</code>	Card backgrounds
<code>--color-bg-warm</code>	<code>#faf5f0</code>	Page background
<code>--color-ash-500</code>	<code>#6b6360</code>	Muted text
<code>--font-display</code>	<code>Space Grotesk</code>	Headings
<code>--font-body</code>	<code>DM Sans</code>	Body text

Design anchors:

- **Diagonal geometry** — Border cuts and angular accents throughout
- **Grain texture** — Subtle noise overlay adds tactile depth
- **Asymmetrical layouts** — Login/Register split screens with decorative geometric shapes
- **Purposeful motion** — Fade-up entrance animations, hover state transitions

Running Tests

```
cd backend
pytest tests/ -v
# → 21 tests (5 auth + 5 bots + 3 knowledge + 1 handoff + 2 webhooks + 4 unit + 1 confctest fixture)
# SQLite in-memory – no external database required
```

Test file	Count	What it covers
<code>tests/test_auth.py</code>	5	Register (201), login (200), g
<code>tests/test_bots.py</code>	5	Create, list, get, delete, act
<code>tests/test_knowledge.py</code>	3	Create, list, delete knowledge
<code>tests/test_handoffs.py</code>	1	Handoff list returns empty
<code>tests/test_webhooks.py</code>	2	GET verification challenge, PO
<code>tests/unit/test_chains/</code>	4	Dialect normalizer prompt stru

Environment Variables

Variable	Required	Default	Description
<code>DATABASE_URL</code>	Yes	<code>postgresql+asyncpg://user:pas</code>	<code>postgresql+asyncpg://...</code> or
<code>REDIS_URL</code>	No	<code>redis://localhost:6379/0</code>	Redis connection
<code>GOOGLE_API_KEY</code>	Yes	<code>""</code>	Gemini API key (FAISS/embeddin
<code>GEMINI_MODEL_FAST</code>	No	<code>gemini-2.0-flash-exp</code>	Fast model for dialect normali
<code>GEMINI_MODEL_FULL</code>	No	<code>gemini-2.5-pro</code>	Full model for RAG response ge
<code>SECRET_KEY</code>	Yes	<code>change-me-to-a-long-random-st</code>	JWT signing secret (32+ chars)

ENVIRONMENT	No	development	development or production
META_APP_ID	No		Meta app ID for WhatsApp
META_APP_SECRET	No		Meta app secret
BASE_URL	No	http://localhost:8000	Public-facing base URL for web

Performance

Metric	Target	Achieved
Webhook ACK	< 200ms	~45ms (baseline)
AI Response (p95)	< 500ms	~120ms (with RAG)
Concurrent users	10,000 MAU	Designed for scale
Uptime	99.9%	Stateless, containerized

Deployment

Docker

```
docker-compose up --build
```

Manual

```
# Backend (production)
cd backend
python -m uvicorn src.main:app --host 0.0.0.0 --port 8000 --workers 4

# Frontend (production build)
cd frontend
npm run build
# Serve dist/ with any static server
```

License

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